

RYDER PONGRACIC

Atlanta, GA

☎ 678-543-4209 ✉ ryderjmp@gmail.com  [linkedin.com/in/rpongracic](https://www.linkedin.com/in/rpongracic)

 github.com/ryderpongracic1  ryderpongracic1.github.io

Education

Georgia Institute of Technology

Jan. 2027 — May 2028

Master of Science in Computer Science | Specialization: Artificial Intelligence

Atlanta, GA

Georgia Institute of Technology

Jun. 2023 — Dec. 2026

Bachelor of Science in Computer Science — GPA: 3.85/4.0

Atlanta, GA

- Concentrations: Intelligence & People (HCI)
- Coursework: ML, AI, Algorithm Design & Analysis, Data Structures, OOP, Computer Org. & Prog., Linear Algebra, Combinatorics, Probability & Statistics, Robotics & Perception

Experience

Amazon

Jun. 2026 — Aug. 2026

Incoming Software Development Engineer Intern | FinTech Org

Arlington, VA

Georgia Tech Bhamla Lab

Aug. 2024 — Dec. 2025

Computational Researcher

Atlanta, GA

- Engineered **MATLAB stochastic simulations** & constructed Bluetooth **Arduino** robots to model & validate control frameworks for pairwise & ternary herding behaviors.
- Designed financial simulations with **LangGraph LLM agents** & **Pandas** to apply herding dynamics to market traders.

Georgia Tech School of Civil Engineering

Jan. 2024 — May 2024

Computer Vision Researcher

Atlanta, GA

- Engineered a custom **High Boost filter mask** & tuned parameters to optimize baseline street crack registration models, driving a **140% increase** in registration quality.
- Architected custom **image interpolation techniques** to salvage failed registrations from a 6-year 3D laser dataset, yielding a **500% increase in recall** & a **263% increase in precision**.

Projects

Distributed Key-Value Store | Go, gRPC, LSM-Trees, Raft, Docker

Apr. 2026 — Present

- Architected a **Golang** distributed key-value store as a strictly consistent **CP System** with a **Consistent Hash Ring** (150 vnodes/node) for balanced **data partitioning & replication**.
- Engineered an **LSM-Tree Storage Engine** with **leveled compaction** (L0→L1, bounded read amplification) & **write-stall backpressure**; added a **32-shard LRU block cache** achieving **99.2% hit rate** on hot-key workloads.
- Hardened the **Write-Ahead Log** via **sync.Pool** buffers, rolling CRC accumulator, & **unsafe** zero-copy key views — reducing allocations **5→2 per append**; torn-write replay stops cleanly on CRC mismatch.
- Implemented full **Raft Consensus** (pre-vote, snapshots, heartbeats) & validated correctness with a **Porcupine linearizability harness** & **Jepsen-style chaos runner**.

Vectorized SQL Query Engine (vexq) | Go, SQL, Columnar Storage

Jan. 2026 — May 2026

- Engineered a complete analytical query pipeline in **Golang**: custom **.vxq columnar format** with per-row-group **zone maps**, dictionary-encoded strings (equality → integer compare), and **CRC32** block integrity.
- Built a **vectorized Volcano-model execution engine** with 1024-row **L1-cached batches**, **selection-vector filters** (zero allocation on hot path), a **Pratt SQL parser**, and predicate-pushdown optimizer — achieving up to **5.3×** over SQLite on TPC-H Q1 at 6M rows.
- Implemented **morsel-driven parallelism** with an **atomic work-stealing scheduler** dispatching row groups across goroutines; lock-free partial **HashAggregate** merge yields an additional **1.1×** speedup on aggregate queries.

CodeScribe AI | Python, FastAPI, Gemini & Groq API, NetworkX

Aug. 2025

- Engineered a hackathon-winning AI-documentation tool with a **FastAPI backend** that parses Python codebases using the **AST module** to map structural context into a **NetworkX** dependency graph.
- Integrated **LLM APIs** to analyze NetworkX graphs, generating highly accurate & context-aware documentation.

Technical Skills

Languages: Python, C/C++, Go, Java, JavaScript, SQL, Cypher, MATLAB, HTML/CSS

Frameworks & Libraries: gRPC, FastAPI, Flask, OpenMP, Pandas, NetworkX, React & React Native, Ionic

Databases & Tools: PostgreSQL, PostGIS, Neo4j, Docker, Git/GitHub, REST APIs

Concepts & AI: Distributed Systems, LSM-Trees, Raft, Vectorized Execution, LLMs, LangGraph, Parallel Processing, CV